

Zachary Stone

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Embedded Software Engineer

Driven embedded engineer seeking to address twenty-first century challenges and increase global welfare through novel technological development.

EDUCATION

B.S. in Electrical Engineering — University of Connecticut, May 2022 — graduated cum laude (3.76)

- Gear Manager, UConn Outing Club Aug. 2021 – May 2022
 - Student Farmer, Spring Valley Student Farm Aug. 2021 – May 2022
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EXPERIENCE

Otis Elevator Company

Farmington, CT

Senior Associate Embedded Software Engineer

Apr. 2025 – Present

Associate Embedded Software Engineer

Apr. 2023 – Apr. 2025

- Develop and maintain real-time, safety-critical motion control software for global high-rise elevator systems in C, built for STM32 ARM microprocessors.
- Innovated test tooling by building a C#/XAML desktop application to simulate elevator system CAN communication, enabling motion control software testing on a minimal hardware testbed.
- Collaborated with field engineering to realize Otis's next generation iPhone-based service tool.
- Developed a multi-lingual C#/XAML desktop application for factory configuration of rope integrity monitoring hardware, eliminating a primary source of new equipment field turn-backs.
- Redesigned elevator controller software communication interfaces to be independent of platform endianness.
- Led full test of motion control software functionality following software port to new computing hardware.

Enviro Power

South Windsor, CT

Electrical Engineering Intern

May 2021 – Sep. 2022

- Designed an ESP32-based data logger PCB utilizing Modbus communication with brownout detection and automatic battery switchover for remote monitoring of boiler health.
- Instituted a PCB quality control procedure and performed quality control assessments on a test stand, reducing unit loss by 8% through hardware fault diagnosis and repair.
- Maintained and improved diagnostic application in C#/XAML for SmartWatt Boiler controller testing.

Power Electronics and Drives Advanced Research Lab

University of Connecticut

Research Assistant

Dec. 2020 – Jun. 2022

- Engineered a variable pole count stator winding for 3 phase induction motors using MotorCAD.
 - Secured a \$5k grant and built a 3 phase variable-pole induction motor, resulting in a 23% power increase above 2000 RPM compared to a non pole-switching design.
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COURSEWORK

Introduction to Robotics

Jan. 2022 – May 2022

- Implemented C++ firmware on an ATmega328P for a robotic grocery cart utilizing SPI and PWM for peripheral control.

Microprocessor Application Lab

Aug. 2021 – Dec. 2021

- Created an autonomous sunshade employing photoresistors, a stepper motor and driver, an I2C joystick, and a UART driven LCD. Control firmware implemented on an ATmega328PB in C.

Systems Lab

Jan. 2021 – May 2021

- Derived physics-based system models and designed PID, PV, and LQR control systems in MATLAB Simulink.
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SKILLS

Languages: C | C++ | C# | Python | ARM Assembly; **Tools:** SPICE | KiCad | FreeCAD | Linux; **Embedded:** CAN | UART | SPI | I2C | JTAG | RTOS; **Benchtop:** Soldering | Oscilloscope | Function Generator;